

Retail Pricing Optimization Analysis

Phase 3: Power BI Dashboard Report

Project Background

This phase focuses on visualizing insights from previous Excel and SQL analysis using Power BI.

Phase 1 identified the profitability threshold using Excel.

Phase 2 validated the findings using SQL and uncovered regional risk patterns.

Phase 3 transforms these findings into interactive business dashboards for easier decision-making.

Objective of Phase 3

The purpose of this phase was to build interactive dashboards that visually present pricing risk patterns and business insights.

The dashboard was designed to answer:

- At what discount level does profitability become negative?
- Which regions are most sensitive to discounting?
- How large is the risky transaction problem?
- How can business users monitor pricing risk visually?

Power BI Skills Applied

This phase included practical use of:

- Data Modeling
- DAX Calculated Columns
- DAX Measures
- Card Visuals
- Bar Charts
- Column Charts
- Line Charts
- Dashboard Layout Design
- Interactive Filtering

Dashboard Pages Created

Page 1: Executive Overview

Purpose

To provide a high-level business summary of pricing performance.

Visuals Included

- Total Sales
- Total Profit
- Profit Margin
- Risky Transactions
- Discount Threshold Analysis

Key Finding

The dashboard confirms that profitability becomes negative after the **21–30% discount range**.

Business Insight

This confirms earlier findings from Excel and SQL.

The company has a clear discount threshold beyond which profitability declines rapidly.

Recommendation

Standard discounts should remain below **20%**.

Page 2: Regional Analysis

Purpose

To compare profitability performance across regions.

Visuals Included

- Regional Profitability Comparison
- Regional Discount Trend Analysis

Key Finding

Regional performance varies significantly.

North Asia shows early profitability decline compared to other regions.

Southeast Asia shows weaker overall profitability.

Business Insight

Different markets respond differently to discounting.

A single discount strategy is not effective across all regions.

Recommendation

Apply region-specific discount rules.

Recommended tighter controls for North Asia.

Page 3: Risk Analysis

Purpose

To evaluate the operational scale of discount-related losses.

Visuals Included

- High-Risk Transaction Count
- Average Loss per Risky Order
- Transaction Distribution by Discount Band
- Category Exposure Analysis

Key Finding

The business processes **11,328 high-risk transactions**.

Average loss per risky transaction is **-71.92**.

Business Insight

The issue is systematic.

Losses are being generated repeatedly through risky pricing decisions.

Recommendation

Implement automated monitoring for transactions above the discount threshold.

Major Insights from Power BI Phase

Insight 1

Dashboard visuals confirmed all previous Excel and SQL findings.

Insight 2

The critical discount threshold remains **20%**

Insight 3

North Asia remains the most discount-sensitive region.

Insight 4

Risky transactions represent a large operational issue.

Insight 5

Power BI enables faster identification of pricing risk patterns.

Final Business Recommendations

1. Maintain company-wide discount ceiling below 20%
2. Apply regional discount customization
3. Monitor risky transactions continuously
4. Review high-loss discount campaigns
5. Use dashboards for pricing decision support

Final Conclusion

Phase 3 successfully transformed analytical findings into business dashboards.

Power BI made the pricing insights easier to interpret and communicate.

The dashboard clearly shows that discounting beyond 20% creates significant profitability risk.

The findings support data-driven pricing optimization.

Project Status

Phase 1: Excel Analysis Completed

Phase 2: SQL Validation Completed

Phase 3: Power BI Dashboard Completed

